

AI Quote Intake

Automated Intelligence for Document Processing

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OUR CHALLENGE

THE PROBLEM

Manual

processing and unscalable workflows

- Brokers submit immense volumes of scanned documents, and contracts
- Insurers must manually scan files to identify risk and fill intake quotes

THE OPPORTUNITY

Automate

extraction to save thousands of hours

- Real insurance data is highly scarce and strictly confidential
- We are only provided with filled documents, no extracted truth.

OUR SOLUTION

82.2%

accuracy + an interactive and auditable app

- End-to-end vision, LLM, and NLP pipeline extracts risk into JSON
- Interactive app lets users verify sources and correct outputs in real time

Insurance Proposal

PROFESSIONAL INDEMNITY INSURANCE — PROPOSAL FORM

Cover Schedule and Risk Disclosure

Catalonia Coastal Engineering Group S.L. | Ref: PI-ES-2024-0718 | 2024-04-01

1. Policy Schedule

(1) Insured Name?	Catalonia Coastal Engineering Group S.L.
(2) Policy Number?	PI-ES-2024-0718
(3) Date Established?	2015-03-15
(4) Policy Start Date?	2024-04-01
(5) Policy End Date?	2025-04-01
(6) Policy Type?	Claims-Made
(7) Cover Trigger?	First Made and Reported
(8) Territory?	Worldwide excluding USA and Canada
(9) Jurisdiction?	Courts of Barcelona, Spain
(10) Limit of Indemnity?	EUR 5,000,000
(11) Excess / Deductible?	EUR 25,000

Blueprint — Details of Parties to be Insured

1. Provide the full legal names and registered addresses of all entities proposed for cover under this policy, including any subsidiaries or associated practices.

Catalonia Coastal Engineering Group S.L., C/ Consell de Cent 301, 08007 Barcelona, Spain

2. Describe the principal nature of business undertaken by the insured parties, including the range of professional disciplines practised and target client sectors.

A&E Consultancy

Clauses, Conditions and Underwriting Notes

Clause 1: Coverage attaches only to professional services declared in this proposal and accepted by underwriting.

Clause 2: Dispute resolution and governing law will follow the jurisdiction stated in the final policy schedule.

Clause 3: Premium calculation assumes accuracy of declared fees and turnover for the insured activities.

Clause 4: Deductible applies each and every claim unless superseded by a specific extension clause.

2. Parties and Risk Overview

Catalonia Coastal Engineering Group S.L. is engaged in A&E Consultancy with primary operations in . Estimated annual turnover is EUR 8,500,000 and professional fees are EUR 7,200,000. Cover is requested on a First Made and Reported basis with retroactive reference date 2019-04-01.

Specialists in coastal protection engineering, port infrastructure design, and maritime structural analysis. Provide advisory services for flood risk management and climate resilience planning.

Type of Business / Profession | A&E Consultancy

Structured Output

1. Policy & Core Information

Policy Number * 100%

PI-2024-00781

Insured Name * 100%

Derwaza Geotechnical and Ground Eng

Date Established 100%

2011-03-14

Policy Wording —

—

Policy Start Date * 100%

2024-07-01

Policy End Date * 100%

2025-06-30

Policy Type * 65%

Project

Cover Trigger * 90%

Claims Made Basis

Requested Retroactive Date

2016-07-01

* 90% Retroactive Limit * 100%

Full Policy Limit

TERRITORY & JURISDICTION

Territory 90%

GCC

Jurisdiction (where claims can be brought) 90%

GCC

2. Risk Information

Limit of Indemnity 97%

Type of Business / Profession 97%



METHOD





EVALUATION

Evaluation

Extraction Score

We evaluate pipeline performance using a weighted utility score defined as

$$\mathcal{S} = \sum_{i=1}^N w_i \cdot \sigma(\hat{y}_i, y_i), \quad \left(\sum_{i=1}^N w_i = 1 \right) \quad \sigma = \begin{cases} 1 & \text{Correct} \\ 0 & \text{Missing} \\ -1 & \text{Incorrect} \end{cases}$$

Discrete Fields: Exact matching criteria.
Text Fields: Levenshtein dist. (threshold ≥ 0.85).
Scale range: $\mathcal{S} \in [-1, 1]$

Extraction Confidence

Confidence is modeled based on answer modifications and retries across the pipeline

$$\text{Conf} = \text{Conf}_{\text{OCR}} \times \text{Conf}_{\text{Span}} \\ \times \text{Canon. Type} \times (0.85)^{N_{\text{teacher retry}}}$$

OCR / Span: DocTR logprobs and mismatch metrics.
Canonicalization Type: Penalizes formatting changes.
Teacher Retries (N): Penalty for every revision.

Results

Results by field type

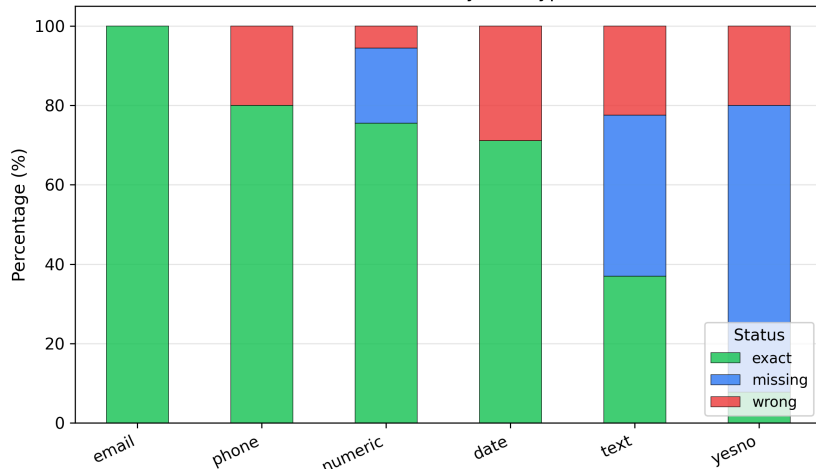
SCORE

$$S = 0.604 \pm 0.3$$

ACCURACY

$$82.3\% \pm 0.42$$

Match Status by field type



Confidence of extracted answer

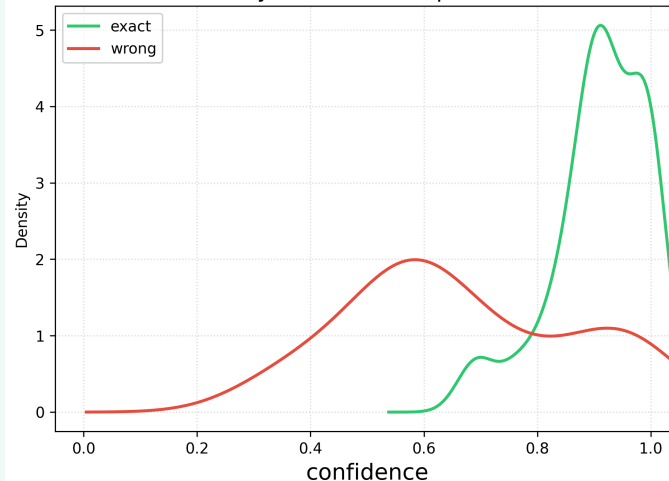
CONFIDENCE THRESHOLD

$$\tau = 0.78$$

HIGH VARIABILITY

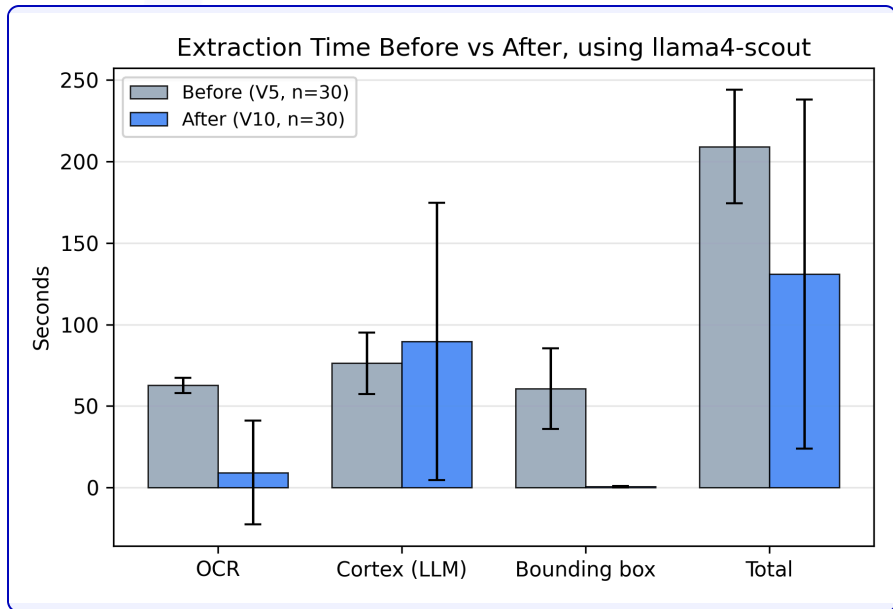
+10 docs

Density of confidence per status



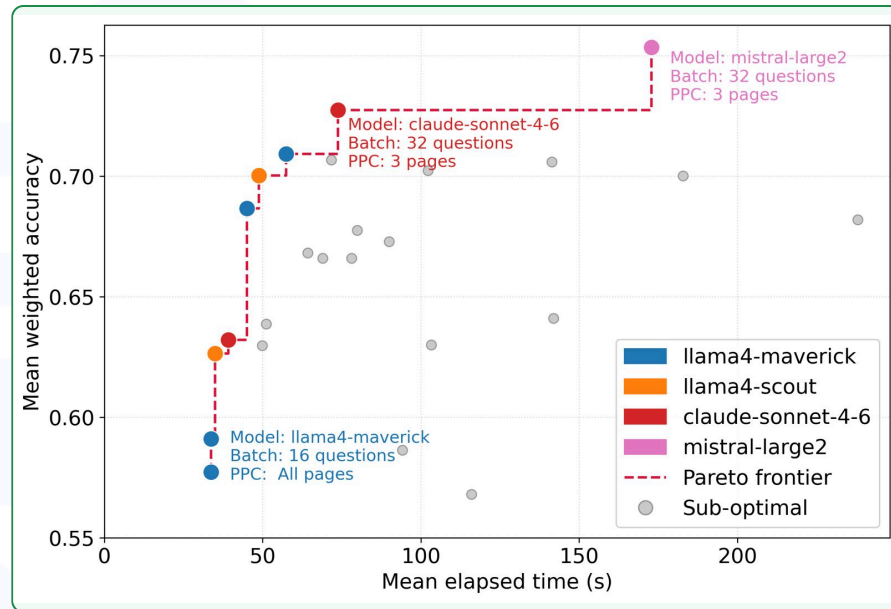
Efficiency

Step runtime (old vs new version)



- **Parallelization** We improve accuracy by parallelizing and reconstructing the fields after OCR.

Model Optimization (Pareto Frontier)



- **Optimal frontier:** 32-question batch limited to 3 pages per query maximizes LLM focus and accuracy.



PLATFORM DEMONSTRATION

Some examples

Platform Demo

PAGE 1

PROFESSIONAL INDEMNITY INSURANCE PROPOSAL FORM SOLICITORS / LAWYERS / NOTARIES

I. GENERAL INFORMATION

- Full name of the proposer: AI QUOTE INTAKE – GROUP
- Address of Head Office: Carreterra de Vic, 25-31, 4-6, Manresa – 08241, (Barcelona) Spain.
- Address of Branch office(s) and name(s) of resident partner(s) covered under this policy:
42 Titas Boulevard, Kandirpar, Cumilla – 3500, Bangladesh
Plabon Shaha
- When was the firm established? 15 of January 2026
- During the past five years has the name of the firm been changed or has any other firm been purchased or any merger or consolidation taken place? No
If so, please give full details:
- Member of association? IREM: Institute of Real state Management
- Details of all practising principals or partners:

Names	Qualifications, dates qualified / total duration of professional	Position held in company and how long
Guillem Masdemont	Ba in Mathematics	1 year
Plabon Shaha	Ba in Computer Science	1 year

1. Policy & Core Information

Policy Number * 86%

RE-2026-00742

Insured Name * 100% (page 1)

AI QUOTE INTAKE – GROUP

Date Established 90%

2020-06-01

Policy Wording 97%

As per the Policy terms and conditions

Policy Start Date * 97%

2026-06-01

Policy End Date * 97%

2027-05-31

Policy Type * 42%

Annual

Cover Trigger * 90%

Claims Made Basis

Requested Retroactive Date * 23%

500000.0

Retroactive Limit * 48%

\$500,000

TERRITORY & JURISDICTION

Territory 62%

Others

Jurisdiction (where claims can be brought) –

–

CONCLUSION



Our solution

We developed a secure pipeline architecture that produces structured JSON files from insurance documents. Everything respecting data privacy.



Accurate on synthetic and manually annotated dataset

We obtain a positive score and high accuracy across different synthetic and manually annotated documents, providing a positive value for the company.



Future Research

We want to focus on formalizing completeness on a generated dataset. Other directions, involve fine-tuning specialized models and improving zero-shot prompting.

Synthetic Data Augmentation

Clean Input (doc001)

Standard digital PDF output with perfect text alignment and no background artifacts.

PROFESSIONAL INDEMNITY INSURANCE — PROPOSAL FORM

Cover Schedule and Risk Disclosure

Derwaza Geotechnical and Ground Engineering Consultants W.L.L. | Ref: PI-2024-00781 | 2024-07-01

1. Policy Schedule

Clauses, Conditions and Underwriting Notes

Clause 1: All amounts stated in this schedule are expressed in Kuwaiti Dinars and all payments of premium, deductible and indemnity shall be settled in that currency unless an alternative is agreed in writing by the insurer prior to settlement.

Clause 2: This policy is issued on a claims-made basis and responds only to claims first made against the insured and notified to the insurer during the period of insurance, provided always that the act, error or omission giving rise to such claim occurred after the retroactive date stated in this schedule.

Clause 3: The insured shall notify the insurer in writing as soon as practicable, and in any event no later than thirty calendar days after the expiry of the policy period, of any circumstance that may give rise to a claim, failing which the insurer reserves the right to decline indemnity in respect of any subsequent claim arising from such circumstance.

Clause 4: The territorial scope of this policy extends to professional services performed by the insured anywhere in the GCC region, provided that any resultant claim is adjudicated before the courts of the State of Kuwait and governed exclusively by Kuwaiti law.

Clause 5: The limit of indemnity stated herein is the maximum aggregate liability of the insurer for all claims and claimable circumstances notified during the policy period, inclusive of defence costs and legal expenses, and shall not be reinstated upon partial exhaustion.

Insured Name	Derwaza Geotechnical and Ground Engineering Consultants W.L.L.
Policy Number	PI-2024-00781
Date Established	2011-03-14
Policy Start Date	2024-07-01
Policy End Date	2025-06-30

Noisy Version (doc001_noisy)

Simulated scan noise: rotation (+/- 2°), Gaussian blur, and brightness variations.

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